Post Quantum Cryptography

Transition Phase I - Hybrid Model

General Notes - Vatsal Patel

Executive Summary

Problem Statement & Objective:
Current Classic Cryptography is NO MORE SECURE!

Current Threat:
Shor's algorithm:
Attacks Public-key Cryptography by efficiently solving integer factorization and discrete logarithm problems on a sufficiently powerful quantum computer.

- Breaks Asymmetric cryptography
- Threatens: RSA, ECC/ECDSA, ECDH

Grover's algorithm:
Speeds up brute-force search roughly from 2^n to 2^n/2.

- Reduces the Security Strength of Symmetric cryptography / Hashes.
- Threatens: SHA, AES(keys)

Q-Day:
It's just a matter of time, by when Quantum Computers will effectively break the current Cryptography.

Existing Solution:
Post Quantum Cryptography

Requirement:
Crypto Agility - Transition to Post Quantum Cryptography

Current Security Posture - Timeline - 1/2

Pre-PQC era / **Today's Requirement**	**Future** Q-Day / PQC onward **Requirement**
• Email / Document Signing and Encryption Encryption - Confidentiality Threat ○ End-to-end Encryption and Authentication Vulnerable against Threat - "Harvest Now, Decrypt Later"	Vulnerable to Harvest Now, Decrypt Later
• Network Security Protocol (TLS, VPN) Key-Establishment ○ Symmetric Key Mechanism is Vulnerable against Threat - "Harvest Now, Decrypt Later"	Defence - Forward Secrecy • Network Security Protocol (TLS, VPN) Authentication ○ Asymmetric Key Mechanism is NOT vulnerable against Threat - "Harvest Now, Decrypt Later"

Current Security Posture - Timeline - 2/2

Pre-PQC era / **Today's Requirement**

- Software Update (features one of any following two)
  - SW verification mechanism SHALL be Updatable, or
  - Verification mechanism SHALL support verification of Quantum-Resistant Signature
- Network Security Protocol (TLS, VPN) Key-Establishment
  - Symmetric Key Mechanism is Vulnerable against Threat - "Harvest Now, Decrypt Later"

Future Q-Day / PQC onward **Requirement**

- Authentication (switch in following both) Implementation and Deployment of
  - Quantum-Resistant Algorithms for Authentication, and
  - Respective PKI Infrastructure
- Network Security Protocol (TLS, VPN) Authentication
  - Asymmetric Key Mechanism is NOT vulnerable against Threat - "Harvest Now, Decrypt Later"

Required to Stay
Future-Proof / Prepared Today

Transition Path

Legacy:
Current Cryptography (CC) - Quantum Vulnerable Cryptography



Transition Phase 1: CC + PQC (Hybrid Solution)
Current Cryptography - Quantum Vulnerable Cryptography
\+ Post Quantum Cryptography - Quantum Resistant Cryptography



Transition Phase 2: PQC only
Post Quantum Cryptography - Quantum Resistant Cryptography

Transition Phase 1: Hybrid Solution

Primary Fact

PQC Solution
Implemented, Deployed or
can be added in Legacy System

→ Then only Hybrid Solution Possible

- New System / Device developed with Post Quantum Cryptographic support can only be Hybrid Solution.
- Legacy System / Device with only Current (Quantum Vulnerable) Cryptography cannot be a Hybrid Solution.
- Hybrid Solution requires both - the Current and Post-Quantum Cryptography to co-exist and be functional.

Transition Phase 1: Hybrid Solution

Valid Cases (Composite Scheme)

The valid hybrid solution requires,
Both - Current and Post-Quantum solutions shall **co-exist in serial order**.

Prospective Threat:
Newly Developed PQC solutions are not sufficiently deployed and proven to be 100% robust against any possible threat.

Solution:
By keeping both the solution is serial order and considering during this transition phase at least one of the two solution is Secure, the System / Device is Secure.

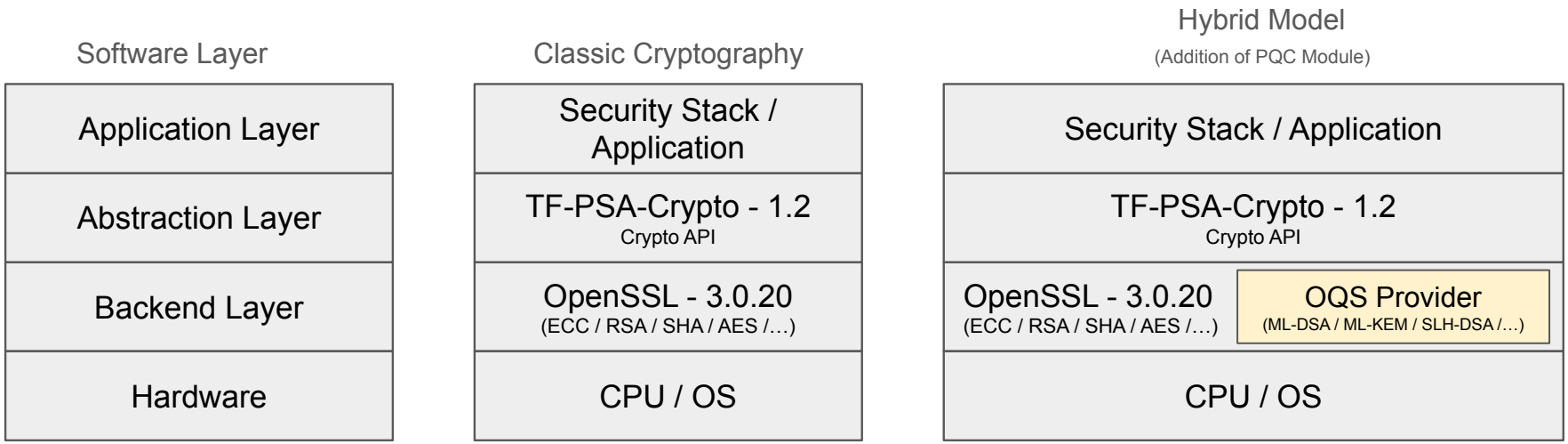
Invalid Cases (Coexistence Scheme)

The cases which looks Hybrid
but NOT a Hybrid Scheme as per requirement criteria

The System / Device implement and deploy both Current and Post-Quantum solutions and security functions are performed by any single solution.

Any technological advancements Hybrid model Where legacy and new system can agree on any single compatible approach from all the available (new or legacy) solutions, become compatible using new or old approach and proceed,
is NOT the case here!!!

Proof Of Concept - Executable Hybrid Model - 1/2



Available Solutions

“OQS-OpenSSL” library
Post Quantum Cryptographic Protocol implementations are available for **real-time use**.

“liboqs” library
Post Quantum Cryptographic Algorithms implementations are available for **research** and **prototyping**.

Proof Of Concept - Executable Hybrid Model - 2/2

Demo

Usage	Classic	Hybrid Model (Today)	PQC (Future)
Digital Signature	ECDSA	ECDSA + ML-DSA	ML-DSA
Key Exchange	ECDH P-256	ECDH P-256 + ML-KEM	ML-KEM

In Progress…

Executable Demo available @ …

Benchmark

Required **additional Resources** and **Performance Impact** to include PQC support

…

In Progress…

Introduced new Security Strength / Category

Security Strength

Historically, the security strength that an algorithm could provide was defined in terms of the amount of work (i.e., the number of operations) that is required to break the algorithm (i.e., an algorithm has s bits of security strength if breaking the algorithm requires 2s operations of some kind, where s = 112, 128, 192, or 256)

Security Category

There are significant uncertainties in estimating the security strengths of post-quantum cryptosystems given the difficulty of accurately predicting the performance characteristics of future quantum computers, such as their cost, speed, and memory size.

Category is defined by a comparatively easy-to-analyze reference Primitive.

Post-Quantum Security Categories

Security Category	Attack Type	Example
1	Key search on a block cipher with a 128-bit key	AES-128
2	Collision search on a 256-bit hash function	SHA-256
3	Key search on a block cipher with a 192-bit key	AES-192
4	Collision search on a 384-bit hash function	SHA3-384
5	Key search on a block cipher with a 256-bit key	AES-256

Appendix - I
Quantum-Vulnerable & Post-Quantum Digital Signature Algorithms

Quantum-Vulnerable Digital Signature Algorithms

Digital Signature Algorithm Family	Parameters	Transition
ECDSA [FIPS186]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035
EdDSA [FIPS186]	≥ 128 bits of security strength	***Disallowed*** after 2035
RSA [FIPS186]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035

Post-Quantum Digital Signature Algorithms

Digital Signature Algorithm Family	Parameter Sets	Security Strength	Security Category
ML-DSA [FIPS204]	ML-DSA-44	128 bits	2
	ML-DSA-65	192 bits	3
	ML-DSA-87	256 bits	5
SLH-DSA [FIPS205]	SLH-DSA-SHA2-128[s/f] SLH-DSA-SHAKE-128[s/f]	128 bits	1
	SLH-DSA-SHA2-192[s/f] SLH-DSA-SHAKE-192[s/f]	192 bits	3
	SLH-DSA-SHA2-256[s/f] SLH-DSA-SHAKE-256[s/f]	256 bits	5
LMS, HSS [SP800208]	With SHA-256/192 With SHAKE256/192	192 bits	3
	With SHA-256 With SHAKE256	256 bits	5
XMSS, XMSS^{MT} [SP800208]	With SHA-256/192 With SHAKE256/192	192 bits	3
	With SHA-256 With SHAKE256	256 bits	5

Appendix - II
Quantum-Vulnerable & Post-Quantum Key-Establishment Scheme

Quantum-Vulnerable Key-Establishment Scheme

Key Establishment Scheme	Parameters	Transition
Finite Field DH and MQV [SP80056A]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035
Elliptic Curve DH and MQC [SP80056A]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035
RSA [SP80056B]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035

Post-Quantum Key-Establishment Scheme

Key Establishment Scheme	Parameter Sets	Security Strength	Security Category
ML-KEM [FIPS203]	ML-KEM-512	128 bits	1
	ML-DSA-768	192 bits	3
	ML-DSA-1024	256 bits	5

Appendix - III
Quantum-Vulnerable & Post-Quantum Block Cipher

Block Cipher	Parameter Sets	Security Strength	Security Category
AES [FIPS197]	AES-128	128 bits	1
	AES-192	192 bits	3
	AES-256	256 bits	5

Appendix - IV
Quantum-Vulnerable & Post-Quantum HASH and XOFs

Hash/XOF Algorithm Family	Variants	Collision Security Strength	Collision Security Category	Preimage Security Strength	Preimage Security Category
SHA-1 [FIPS180]	SHA-1	80 bits	< 1	160 bits	1
SHA-2 [FIPS180]	SHA-224 SHA-512/224	112 bits	< 1	224 bits	3
	SHA-256 SHA-512/256	128 bits	2	256 bits	5
	SHA-384	192 bits	4	384 bits	5
	SHA-512	256 bits	5	512 bits	5
SHA-3 [FIPS202]	SHA3-224	112 bits	< 1	224 bits	3
	SHA3-256	128 bits	2	256 bits	5
	SHAKE128	128 bits	2	128 bits	2
	SHA3-384	192 bits	4	384 bits	5
	SHA3-512	256 bits	5	512 bits	5
	SHAKE256	256 bits	5	512 bits	5

Reference

https://openquantumsafe.org/

https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf